REAL-TIME MULTIMODAL DRIVER STRESS MONITORING AND ADAPTIVE FEEDBACK VIA BIOSIGNALS AND CONVERSATIONAL AI

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BONAFIDE CERTIFICATE

Certified that this report, titled "**REAL-TIME MULTIMODAL STRESS MONITORING AND ADAPTIVE FEEDBACK VIA BIOSIGNALS AND CONVERSATIONAL AI**" is the bonafide work of **C Padmapriya (3122 22 5002 089) and U Pranaav (3122 22 5002 093)** who carried out the work under my supervision.

Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Driver stress is a critical factor affecting road safety, reaction time, and cognitive performance. Traditional monitoring methods based on visual cues or single-channel physiological signals often fail in real-world driving due to noise, variability, and limited generalization. This project proposes an AI-enabled multimodal driver stress monitoring system designed for reliable, real-time evaluation of a driver's physiological and emotional state.

The system uses ECG, EMG, respiration, heart rate, and GSR signals collected during driving. After evaluating ensemble ML models such as Random Forest, Gradient Boosting, SVM, and Stacking, their limitations in capturing complex temporal patterns became evident. To overcome this, advanced deep learning architectures including CNN-BiLSTM-Attention, Parallel CNN-LSTM, ResNet1D-LSTM, Inception1D-LSTM, and CNN-GRU-Attention were developed, leveraging temporal structure and multi-scale features for significantly improved stress recognition.

The framework is designed to support real-time adaptive feedback, and future work will incorporate a conversational AI module for context-aware driver assistance.

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LIST OF ABBREVIATIONS

NO.	ACRONYM	ABBREVIATIONS
1.	LSTM	Long Short Term Memory
2.	CNN	Convolutional Neural Network
3.	SVM	Support Vector Machine
4.	RNN	Recurrent Neural Network
5.	ResNet	Residual Networks
6.	RF	Random Forest
7.	GB	Gradient Boosting
8.	GRU	Gated Recurrent Unit
9.	ML	Machine Learning

CHAPTER 1
INTRODUCTION

1.1 Overview

Driver stress is a major factor affecting cognitive performance, decision-making ability, and overall road safety. High stress levels can impair reaction time, reduce situational awareness, and significantly increase the probability of road accidents. Traditional approaches to driver monitoring have relied primarily on visual cues such as facial expressions or gaze tracking, or on single physiological signals such as heart rate or skin conductance. However, such unimodal systems often fail under real-world conditions due to motion artifacts, variable lighting, occlusions, and inter-individual differences in stress expression.

To overcome these limitations, this project proposes a real-time multimodal driver stress detection framework that leverages complementary biosignals to produce a more reliable and holistic assessment of the driver's physiological and emotional state. The system integrates five major physiological modalities, Electrocardiogram (ECG), Electromyogram (EMG), Respiration (RESP), Heart Rate (HR), and Galvanic Skin Response (GSR) from both hand and foot sensors, collected under naturalistic driving conditions across 17 drivers, totaling more than 22 hours of physiological recordings.

The project aims to achieve three core objectives:

1. **Multimodal Stress Detection:** Identify low, medium, and high driver stress levels by analyzing synchronized cardiovascular, muscular, respiratory, and electrodermal responses.
2. **Real-Time Stress Prediction:** Build and evaluate advanced deep learning architectures, including CNN–BiLSTM–Attention, Parallel CNN-LSTM, ResNet1D-LSTM, Inception1D-LSTM, and CNN–GRU–Attention, to

determine the most effective model for offline evaluation and real-time deployment.

3. **Adaptive Conversational Feedback:** Lay the foundation for integrating a lightweight LLM-based conversational assistant capable of providing empathetic, context-aware interventions during stressful driving episodes.

Traditional machine learning models such as Random Forest, Gradient Boosting, SVM, and soft-voting ensemble classifiers were first evaluated as baselines. Although these conventional models delivered reasonable performance, they were limited in capturing long-term temporal variations and multimodal dependencies. Advanced deep learning architectures, especially CNN-BiLSTM-Attention and CNN-GRU-Attention, offered substantial improvements by jointly learning local temporal features, recurrent dependencies, and attention-based weighting of significant physiological patterns.

Among all tested models, CNN-BiLSTM-Attention achieved the highest offline accuracy on fixed-length windows, whereas the CNN-GRU-Attention model demonstrated superior efficiency and practicality for real-time deployment, owing to its faster inference speed and significantly lighter parameter count.

1.2 Motivation

The increasing shift toward intelligent transportation systems demands reliable in-vehicle monitoring solutions capable of assessing driver safety in real time. Stress-induced impairment is a critical yet under-addressed challenge that cannot be mitigated solely through visual monitoring. A robust physiological stress detection system offers an essential layer of safety by capturing subtle internal cues that cameras overlook.

The motivation behind this project is driven by several factors:

- **Enhancing Road Safety:** Early detection of elevated stress allows intelligent systems to intervene proactively, reducing the likelihood of accidents.

- **Overcoming Unimodal Limitations:** Single signals often fail due to noise, artifacts, or context-specific variability; multimodal integration creates a more stable and generalizable system.
- **Human-Centered Intervention:** With the emergence of conversational AI, cars can now deliver supportive feedback, calming prompts, or context-aware recommendations in stressful situations.
- **Advancing Emotion-Aware AI:** Physiological signals offer a pathway toward more empathetic and emotionally intelligent computing systems in automotive, healthcare, and industrial domains.

This project aims to bridge the gap between driver physiology, machine learning, and real-time adaptive feedback to create safer and more emotionally supportive intelligent vehicles.

1.3 Challenges

Developing a robust multimodal stress detection system involves several critical challenges:

1. **Physiological Noise and Motion Artifacts:** Driving environments introduce significant noise due to movement, respiration changes, steering effort, and road vibrations, making clean signal extraction difficult.
2. **Inter-Individual Variability:** Physiological stress responses differ widely between individuals, requiring models that generalize across drivers.
3. **Multimodal Synchronization:** ECG, EMG, RESP, HR, and GSR have different sampling rates and response characteristics, requiring careful preprocessing and alignment.
4. **Data Scarcity:** High-quality, annotated physiological datasets for real-world driving are limited, making model training and evaluation challenging.
5. **Real-Time Constraints:** While complex architectures (e.g., BiLSTM) perform well offline, they require access to future time windows, reducing suitability for

real-time deployment where only past data is available.

6. **Computational Efficiency:** In-vehicle systems require lightweight models (like GRU-based architectures) with low latency and high throughput.

Overcoming these challenges requires precise signal engineering, carefully selected deep learning architectures, and a robust multimodal design strategy.

1.4 Roadmap

This study follows a structured and systematic approach to develop a real-time multimodal driver stress monitoring and adaptive feedback system. The research is organized into five chapters, each addressing an essential component of the project:

1. **Introduction**: This chapter establishes the importance of real-time driver stress detection and its role in enhancing road safety and human–vehicle interaction. It outlines the project's objectives, motivations, and the challenges associated with analyzing multimodal physiological signals in dynamic driving environments.
2. **Literary Survey**: This section presents an overview of existing research on physiological stress recognition. It examines classical machine learning methods, multimodal fusion techniques, deep temporal neural networks, and in-vehicle monitoring technologies, providing insight into their capabilities and limitations.
3. **Methodology**: This chapter describes the end-to-end workflow of the system. It includes data acquisition procedures, signal preprocessing techniques, multimodal feature extraction strategies, stress-level labeling, and the design of the evaluation pipeline. Block diagrams and workflow illustrations are provided to clarify each stage of the system.
4. **System Design and Architecture**: This section details the implementation of the proposed models, covering both traditional ensemble-based approaches and GPU-optimized deep learning architectures. It explains the

structure and design rationale behind CNN–BiLSTM–Attention, Parallel CNN-LSTM, ResNet1D-LSTM, Inception1D-LSTM, and CNN–GRU–Attention models, along with their architectural characteristics and computational profiles.

5. **Results and Discussions**: This chapter presents the experimental setup and evaluation of the implemented models. Performance metrics, confusion matrices, and comparative analyses are included to highlight the behaviour of the system under various configurations. Observations related to generalization, multimodal contributions, and real-time suitability are also discussed.
6. **Conclusion and Future Work:** The final chapter summarizes the research contributions and outlines future directions, including the integration of fuzzy logic for adaptive decision-making, multimodal transformer-based fusion, and real-time conversational AI feedback for driver assistance.

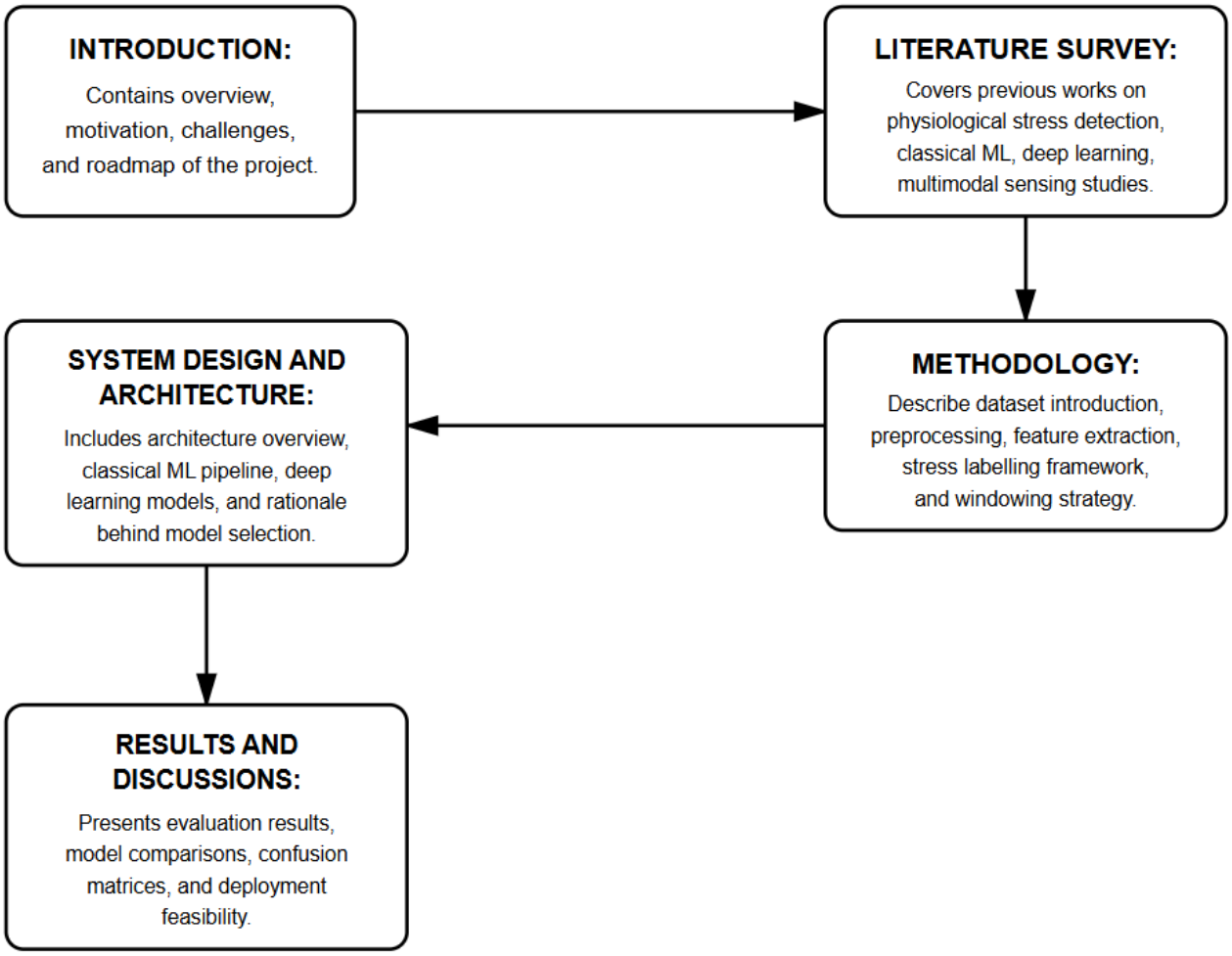


Fig 1.1: Roadmap of the project flow

CHAPTER 2

LITERATURE SURVEY

Healey and Picard (2000) conducted one of the earliest studies on detecting driver stress using physiological signals. Their system combined ECG, EMG, respiration, and skin conductance to classify stress levels during real-world driving. Using multilayer physiological analysis and fixed 5-minute signal windows, the study achieved high classification performance across low, medium, and high stress segments. The research established the feasibility of stress-aware vehicles capable of adapting to driver state, but lacked real-time inference and consistent segment-level labeling frameworks [1].

Vargas-Lopez et al. (2019) proposed an explainable machine learning approach that used statistical time-domain features extracted from EMG signals and trained Support Vector Machine (SVM) classifiers with interpretable kernel functions. The study demonstrated strong predictive capability for stress detection in controlled driving-like environments. EMG was noted for its robustness against factors such as skin moisture and cardiac variations. However, the single-modality design limited generalization to complex and noisy real-world scenarios, and real-time deployment was not validated [2].

Healey et al. (2005) expanded on earlier work by developing wearable and in-vehicle multimodal affect recognition systems. Using over 36 hours of data from ten drivers across laboratory, ambulatory, and automotive setups, the study evaluated ECG, EMG, GSR, and respiration under varied conditions. High recognition accuracy was achieved through multimodal fusion, highlighting the benefits of combining multiple physiological sources. Nonetheless, motion artifacts, individual variability, and limited long-term sensor stability posed significant challenges [3].

Lopez-Martinez, El-Haouij, and Picard (2018) introduced a multi-view, multi-task learning (MT-MKL) framework for driver stress detection using electrodermal activity (EDA) and heart rate signals. The system used spectral clustering to group driving patterns and applied kernel-based multi-task models across multiple real-world datasets including MIT DriveDB, HciLab, and AffectiveROAD. The approach improved stress classification consistency and highlighted the importance of personalization. However, the study only addressed binary stress classification (high vs. low), and did not incorporate additional modalities such as EMG, respiration, or facial cues [4].

Zontone et al. (2019) analyzed stress under manual and autonomous driving conditions using skin potential response (SPR) and ECG recordings. A Support Vector Machine model was developed using SPR features extracted every 15 seconds. The paper concluded that manual driving triggers more frequent stress responses, whereas autonomous driving results in fewer but more intense peaks. Despite the insights, the small sample size, fixed task-based labeling, and exclusion of ECG from ML classification limited the general applicability of the system [5].

Chen et al. (2020) proposed a multimodal driving stress detection model integrating ECG, respiration, and GSR using a combination of time-domain, frequency-domain, and wavelet-based features. Dimensionality reduction was performed using Principal Component Analysis (PCA) and Sparse Bayesian Learning (SBL), followed by kernel classifiers for stress-level prediction. The framework demonstrated reliable stress tracking in real-world traffic transitions. However, scalability remained restricted due to small sample size, and achieving seamless real-time deployment posed practical challenges [6].

CHAPTER 3

METHODOLOGY

3.1 Dataset Introduction

The dataset used in this study consists of multimodal physiological recordings collected from 17 drivers during approximately 22 hours of naturalistic driving. The recordings were captured across varying traffic conditions including city congestion, highways, and resting segments to ensure diverse stress responses.

The dataset includes five key physiological modalities:

- **Electrocardiogram (ECG)** – cardiac electrical activity and R-peak dynamics
- **Heart Rate (HR)** – beat-to-beat variability
- **Electromyogram (EMG)** – muscular tension associated with cognitive workload
- **Respiration (RESP)** – breathing depth, rhythm, and irregularities
- **Galvanic Skin Response (GSR: hand + foot)** – electrodermal stress reactivity

Each recording contains realistic physiological challenges such as motion artifacts, sensor drift, respiration irregularities, and abrupt stress spikes typical of real driving scenarios.

To standardize the dataset for machine learning, all signals were resampled to 100 Hz and segmented into **5-minute windows**, forming the base dataset used for traditional ML and deep learning architectures.

3.2 Data Preprocessing

A comprehensive preprocessing pipeline was implemented to ensure clean, consistent, and synchronized physiological signals across all subjects and modalities. Since raw biosignals often contain motion artifacts, baseline drifts, sensor-induced distortions, and sampling inconsistencies, each modality was handled with a modality-specific filtering, correction, and normalization strategy. The objective of this preprocessing stage was to prepare high-quality, time-aligned signals suitable for multimodal feature extraction and deep temporal modeling.

3.2.1 Filtering and Noise Removal

Each physiological modality exhibits distinct frequency characteristics; therefore, dedicated filters were selected to isolate meaningful physiological information while removing noise and artifacts.

Table 3.1: Preprocessing Filters Applied to Multimodal Physiological Signals

Signal	Filter Applied	Purpose
ECG	Band-pass (0.5–40 Hz)	Remove baseline wander & high-frequency noise
EMG	Band-pass (20–450 Hz)	Isolate muscular activity + artifacts removal
GSR	Low-pass (< 1 Hz)	Extract tonic phasic components
Respiration	Band-pass (0.1–0.8 Hz)	Preserve breathing cycles (6–48 BPM)

Artifact Management

Beyond frequency filtering, several additional steps were performed to mitigate sensor imperfections:

- **Motion Artifact Suppression:** Sudden discontinuities were corrected using median smoothing and adaptive filtering.
- **Signal Interpolation:** Small gaps due to sensor dropout were reconstructed using cubic spline interpolation to preserve waveform continuity.
- **Outlier Removal:** Segments containing physiologically implausible spikes (e.g., sudden HR > 200 BPM) were detected using z-score thresholds and replaced with interpolated values.
- **Drift Correction:** Particularly for GSR, slow tonic drift was corrected using detrending filters to maintain stable baselines.

These steps ensured that all signals retained physiological meaning while minimizing noise contributions.

3.2.2 Resampling and Normalization

Since the physiological modalities came from different sensors with varying native sampling rates, uniform temporal alignment was crucial.

Resampling

All signals were resampled to a common frequency of 100 Hz using polyphase filtering. This ensured:

- Consistent sampling resolution across modalities
- Synchronization for multimodal fusion
- Compatibility with convolutional and recurrent deep learning models

Segmentation

Signals were then segmented into 300-second (5-minute) windows, as this duration offers:

- Sufficient physiological context for HRV, GSR peak dynamics, and respiratory variability
- Compatibility with stress-level definitions from traffic conditions
- Stable statistical feature extraction

Normalization

For each window:

- Z-score normalization was applied to remove inter-subject differences.
- Min–Max scaling was used for GSR to handle individual conductivity variations.
- Each segment was validated for continuity before being passed to feature extraction.

This resulted in stable, uniform input representations across subjects and modalities.

3.3 Feature Extraction

Feature extraction was performed separately for each modality, considering its physiological relevance to stress. The goal was to capture both short-term and long-term variations associated with sympathetic and parasympathetic nervous system activity.

Each 5-minute window underwent feature extraction using domain-appropriate algorithms, with the resulting features concatenated into a comprehensive multimodal feature vector.

3.3.1 ECG and HRV-Based Features

ECG signals were processed using robust R-peak detection methods. From the R–R interval series, standard HRV features were computed:

- **Mean HR** – average heart rate within the window
- **SDNN** – overall HRV reflecting autonomic balance
- **RMSSD** – parasympathetic activation and short-term HRV
- **pNN50** – percentage of R–R pairs differing by >50 ms
- **LF/HF ratio** – measure of sympathetic dominance
- **RR interval statistics** – distribution measures (mean, variance, skewness)

These metrics capture known physiological correlates of stress, including increased sympathetic activation and reduced vagal tone.

3.3.2 EMG Features

EMG reflects muscle tension associated with cognitive workload and grip force on the steering wheel.

Extracted features included:

- **RMS energy** – overall muscular activation
- **Zero crossing rate** – rapid fluctuations linked to micro-tension
- **Skewness and kurtosis** – waveform shape abnormalities during stress
- **Signal energy** – cumulative muscular activation

- **Standard deviation** – variability in tension patterns

These features help assess how stress influences muscular engagement during driving.

3.3.3 Respiration Features

Respiration changes under stress, typically showing faster, irregular, or shallow breathing.

Features included:

- Breathing rate (BPM)
- Variance in inter-breath intervals
- Spectral power in the respiration band
- Peak count and prominence
- Inspiration - expiration duration ratio

These features capture stress-induced changes such as hyperventilation and irregular respiratory patterns.

3.3.4 GSR Features

GSR is one of the most sensitive physiological indicators of emotional stress. Separate features were extracted for hand and foot sensors:

- **Mean skin conductance** – tonic arousal baseline
- **Number of phasic peaks** – rapid stress reactions
- **SCR amplitude distribution** – strength of stress responses
- **Tonic slope** – gradual arousal drift
- **Signal variability** – overall arousal dynamics

GSR reliably captures emotional valence and intensity changes produced by road events.

3.4 Stress Labeling Framework

Stress levels were categorized into Low, Medium, and High, based on a combination of contextual and physiological indicators.

Labeling Criteria

1. **Driving Task Intensity**
    - Rest segments → Low stress
    - Highway segments → Low/Medium stress
    - City congestion, overtaking, braking → Medium/High stress

2. **Physiological Patterns**
    - Sudden rises in HR, GSR peaks → Higher stress
    - Increased EMG tension during maneuvering → Medium/High stress

3. **Rule-Based Thresholds**
    - HR above personalized baselines
    - GSR peak frequency above threshold
    - EMG activity spikes during high workload

Each 5-minute window was assigned a stress label based on a weighted combination of the above rules, ensuring consistent, reproducible segmentation across all subjects.

3.5 Dataset Preparation and Window Structuring

After preprocessing and labeling, the dataset was prepared for machine learning and deep learning pipelines.

Steps Performed

- Feature vectors for each window were saved in structured CSV format for classical ML.
- Sequence tensors were constructed by grouping 10 consecutive windows, enabling temporal modeling in LSTMs and GRUs.
- StandardScaler was used to normalize features across subjects for ML models.
- Time-series tensors were kept in their raw-filtered form for CNN, LSTM, and hybrid models.

This resulted in a complete, multimodal dataset suitable for both traditional and advanced neural architectures.

3.6 Problem Description

The goal of this study is to develop a reliable driver stress detection system using multimodal physiological signals (ECG, EMG, GSR, and Respiration). Since stress affects both short-term signal dynamics and long-term physiological patterns, the methodology was divided into two major objectives:

(1) Classical machine learning for feature-based stress analysis, and
(2) Deep learning for modeling temporal dependencies.

The first objective explores whether carefully engineered multimodal features such as HRV indices, EMG activity descriptors, respiration variability, and GSR tonic/phasic

markers are sufficient to classify stress levels. The second objective investigates advanced deep neural architectures that can capture sequential physiological changes, which are essential when stress evolves gradually during real-world driving.

3.6.1 Objective 1: Classical Machine Learning Stress Classification

3.6.1.1 Trial 1: Ensemble ML Models (RF + Gradient Boosting + SVM)

In the first trial, a soft-voting ensemble classifier was developed by combining Random Forest (RF), Gradient Boosting (GB), and a Radial Basis SVM. This combination was chosen to leverage the complementary strengths of:

- RF for handling nonlinearities and feature interactions,
- GB for modeling subtle feature patterns, and
- SVM for strong margin-based separation between stress levels.

A complete preprocessing pipeline was constructed before model training:

1. Missing-value imputation using median statistics.
2. Outlier suppression using a 1st–99th percentile clipping function.
3. RobustScaler for scaling in the presence of heavy-tailed distributions.
4. SMOTE-based oversampling to correct class imbalance across low, medium, and high stress categories.

The soft-voting ensemble aggregated prediction probabilities from all three classifiers to produce the final output. Performance evaluation was conducted using accuracy, F1-score, and confusion matrix analysis.

While this model achieved moderate accuracy, it struggled with:

- Temporal transitions between stress states,
- Subtle physiological variations during medium-stress conditions, and
- Overlapping feature distributions across classes.

This indicated that handcrafted features, even when combined with strong classical learners, were insufficient to fully capture the sequential nature of biosignals.

3.6.1.2 Trial 2: Stacking Ensemble (LightGBM + CatBoost / Random Forest → Logistic Regression)

To enhance predictive power, a more advanced stacking ensemble was implemented in Trial 2. This model used powerful gradient-boosting algorithms as base learners:

- **LightGBM** (mandatory base learner),
- **CatBoost** (when available) or fallback Random Forest,
- A **Logistic Regression** meta-classifier trained on base learner outputs.

This architecture was selected to capture richer nonlinear dependencies in the multimodal feature space.

Key preprocessing steps included:

- Median imputation,
- Outlier clipping,
- RobustScaler normalization,
- SMOTE augmentation,
- Removal of constant or low-variance features,
- Stratified cross-validation to ensure balanced evaluation.

Compared to the soft-voting model, the stacking ensemble offered:

- Stronger handling of heterogeneous biosignal features,

- Better discrimination between stress levels,
- Improved generalization through cross-validated meta-learning.

LightGBM's tree-based feature importances were also analyzed to understand which physiological features such as HRV indices, GSR phasic peaks, or EMG RMS, contributed most to stress prediction.

Although the stacking model performed better than Trial 1, its performance still plateaued due to the inability of classical models to learn temporal relationships leading to the shift toward sequence-learning deep neural networks in Objective 2.

3.6.2 Objective 2: Deep Learning for Sequence Modeling

To overcome the limitations of feature-based classifiers, deep learning architectures were introduced to learn patterns directly from temporally ordered sequences. Each model was trained on 10-window (50-min) input sequences using FP16 mixed precision on an RTX 4060 GPU for efficiency.

The models were designed to:

- learn high-level patterns directly from raw feature sequences
- capture short-term and long-term stress variations
- improve sensitivity to transitions between stress states
- enable real-time deployment through optimized architectures

3.6.2.1 Trial 1: CNN-BiLSTM-Attention

This architecture integrates three major components:

1. **CNN Layers**
    - capture morphological patterns in HRV fluctuations, EMG bursts, GSR

peaks, and respiration cycles
    - learn local temporal filters and remove noise
2. **BiLSTM Layers**
    - process physiological data in both forward and backward directions
    - learn long-term dependencies across the full 5-minute segments
3. **Attention Module**
    - assigns higher weight to stress-informative timesteps
    - improves interpretability by highlighting peak segments

This model provided the best offline classification performance, benefiting from rich temporal memory and bidirectional context.

3.6.2.2 Trial 2: Parallel CNN–LSTM (Multi-Scale Dual Stream)

A dual-stream architecture was created to learn multi-scale temporal features:

- **Stream 1:** 3×1 convolution → LSTM
- **Stream 2:** 5×1 convolution → LSTM

Outputs were fused to combine short-term and long-term temporal granularity. This approach captured differences between:

- rapid EMG bursts
- slower respiratory variations
- mid-range GSR peaks

The architecture exhibited strong robustness but increased computational complexity.

3.6.2.3 Trial 3: ResNet1D–LSTM

This model introduced:

- Residual connections (improving gradient flow)
- Deep hierarchical temporal representations
- Two-tier LSTM layers for long-term sequence modeling

ResNet1D helped extract cleaner features from noisy biosignals and reduced vanishing-gradient issues in deeper settings.

3.6.2.4 Trial 4: Inception1D–LSTM

Inspired by Google's Inception modules, each block used:

- 1×1, 3×1, 5×1 convolutions
- parallel MaxPooling
- concatenated multi-scale outputs

This multi-scale design enabled modeling stress patterns emerging at different physiological timescales. When combined with LSTM layers, the architecture could track gradual stress buildup over extended periods.

3.6.2.5 Trial 5: CNN–GRU–Attention

A lightweight alternative designed for real-time deployment.

- GRU layers replaced LSTMs (same performance, fewer parameters)
- Faster training and inference

- Suitable for on-board vehicle computing (Raspberry Pi / Jetson Nano)
- Attention layer enhanced sensitivity to stress peak segments

Although smaller, the model offered competitive accuracy and the best computational efficiency.

Table 3.2: Stress Level Class Definitions

Class ID	Class Name	Description
0	low	Low level of stress
1	medium	Medium level of stress
2	high	High level of stress

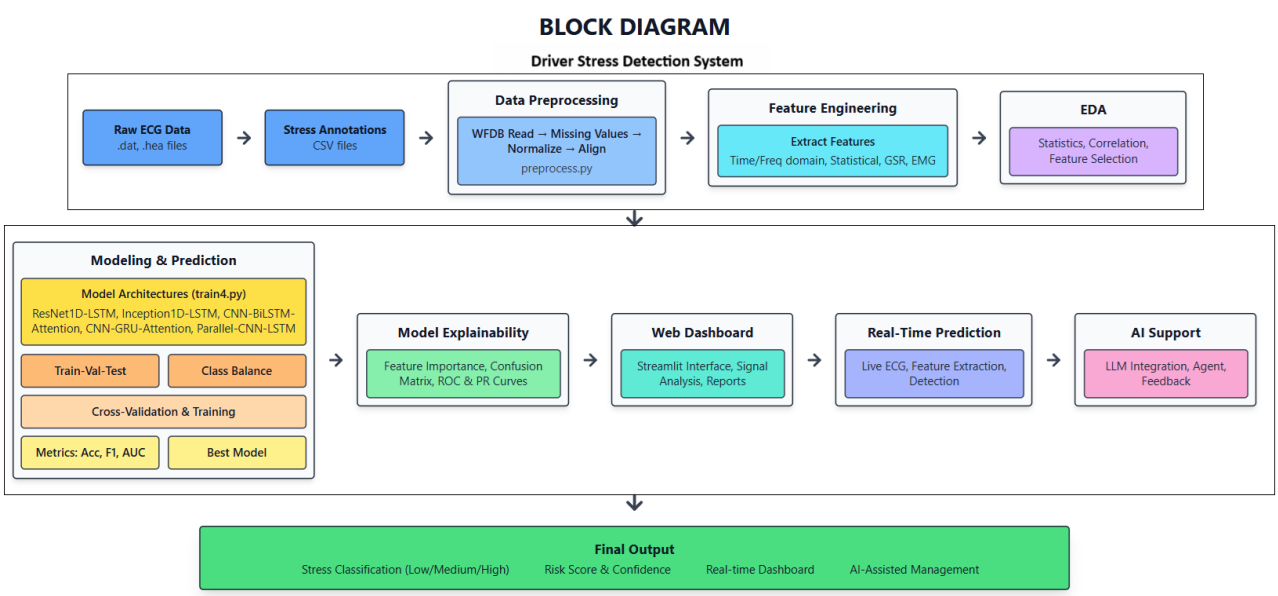


Fig 3.1 : Work Flow Diagram of Objective 2

CHAPTER 4

SYSTEM DESIGN AND ARCHITECTURE

This chapter presents a comprehensive description of the algorithms, model architectures, and computational strategies used in developing the multimodal driver stress detection system. It explains how classical machine learning and advanced deep learning models were incorporated, the rationale behind their selection, mathematical foundations where applicable, and the overall workflow followed to address the challenges of physiological stress modeling.

4.1 Overview of the System Architecture

The overall system integrates multiple stages like signal acquisition, preprocessing, multimodal feature extraction, temporal sequence modeling, model training, and classification. The architecture is organized into two major pipelines:

1. **Classical Machine Learning Pipeline**
   Uses engineered physiological features from ECG, EMG, GSR, and Respiration.
2. **Deep Learning Pipeline**
   Operates directly on sequential 1D windows and captures long-range temporal dependencies.

Both pipelines are evaluated to understand the potential trade-offs between interpretability, complexity, and performance.

4.2 Classical Machine Learning Algorithms

Classical algorithms serve as the first baseline for stress classification using handcrafted features. They evaluate whether statistical physiological markers alone are sufficient to detect stress.

4.2.1 Soft Voting Ensemble Classifier

A soft-voting ensemble combines predictions from multiple heterogeneous classifiers, in this case, Random Forest, Gradient Boosting, and Support Vector Machine.
Each model outputs class probabilities, which are averaged to produce a final decision.

A. Random Forest (RF)

Random Forest constructs multiple decision trees using randomly sampled data and features.
The final prediction is obtained through majority voting across trees.
The decision rule is:

$$\hat{y} = \arg\max_{k} \sum_{t=1}^{T} \mathbb{I}(h_t(x) = k) \tag{4.1}$$

B. Gradient Boosting (GB)

Gradient Boosting builds an additive model where each weak learner corrects the residual error of previous learners.

The functional update rule is:

$$F_m(x) = F_{m-1}(x) + \nu\, h_m(x) \tag{4.2}$$

C. Support Vector Machine (RBF-SVM)

SVM finds a maximum-margin decision boundary.

The RBF (Gaussian) kernel maps nonlinear physiological patterns into an infinite-dimensional feature space:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \tag{4.3}$$

Soft Voting Prediction Rule

Soft voting averages the predicted class probabilities from all base learners:

$$\hat{y} = \arg\max_k \sum_m p_m(y = k \mid x) \tag{4.4}$$

Working Steps

- Train each classifier independently using multimodal feature vectors.
- Aggregate probability scores from each classifier.
- Select the final class label using the highest average probability.

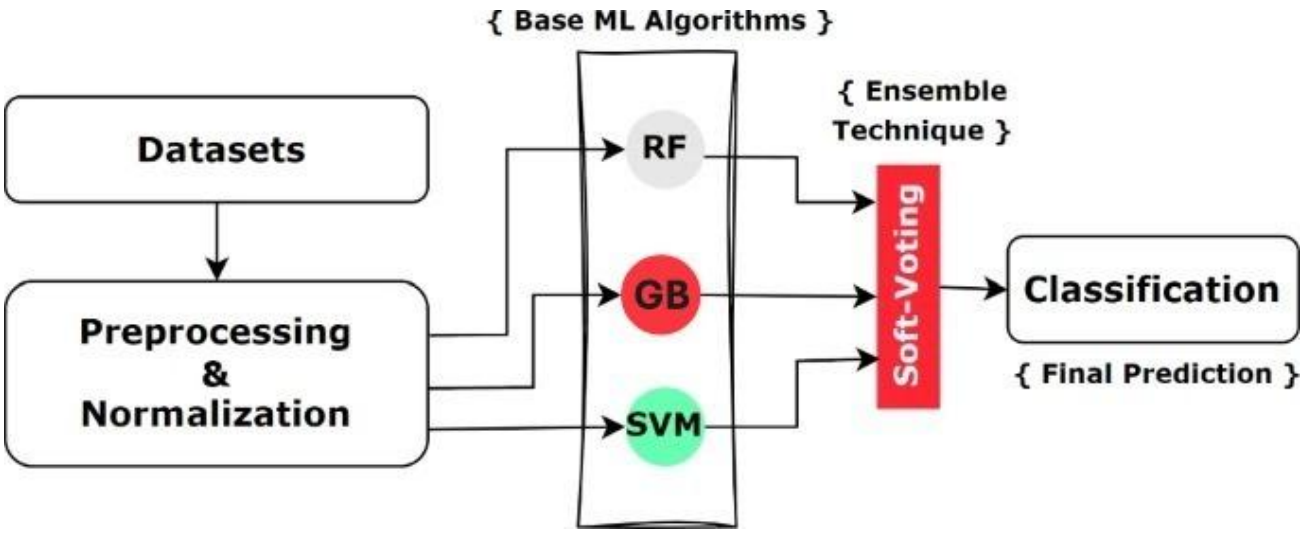


Fig 4.1 : Ensemble Machine Learning Model

Strengths

- Robust to overfitting due to diversity of base learners.
- Improves stability of predictions for noisy physiological signals.

Limitations

- Does not capture temporal evolution within signals.
- Accuracy saturates when feature separability is low.

4.2.2 Stacking Ensemble (LightGBM + CatBoost + Logistic Regression)

Stacking combines high-capacity tree-based learners to exploit complex nonlinear relationships in the features.

Architecture

- **Level 1 models:** LightGBM and CatBoost trained on multimodal features.

- **Meta-learner:** Logistic Regression trained on Level 1 predictions.

A. LightGBM Classifier

A histogram-based gradient boosting technique:

$$\min_f \sum_{i=1}^{N} l(y_i, f(x_i)) + \Omega(f) \tag{4.5}$$

LGBM was chosen for:

- fast training
- strong performance on tabular stress features
- built-in handling of non-linear interactions

B. CatBoost Classifier

Handles categorical interactions and reduces overfitting through Ordered Boosting.

C. Meta-Learner: Logistic Regression

Receives predictions from base models and computes final decision:

$$P(y = 1) = \frac{1}{1 + \exp(-w^T x)} \tag{4.6}$$

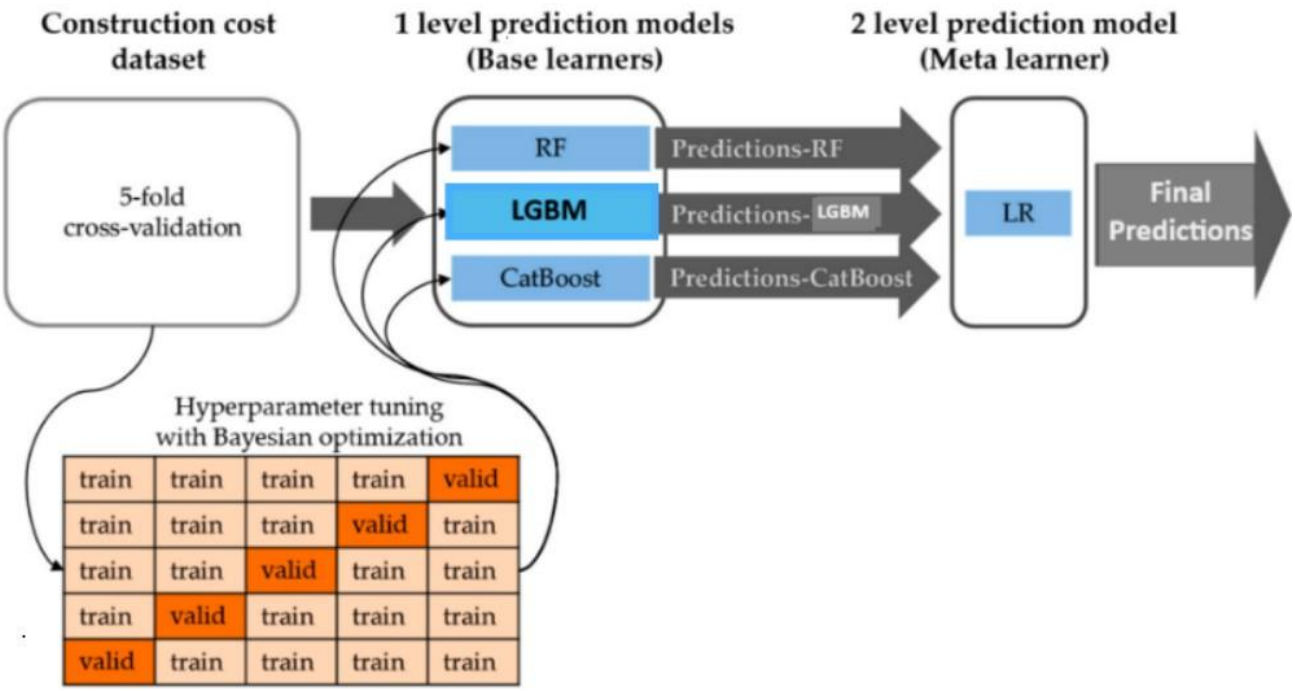


Fig 4.2 : Stacking Ensemble Architecture

Advantages

- Excellent performance on tabular physiological datasets.
- Captures complex interactions without extensive preprocessing.

Limitations

- Still fundamentally static, no temporal sequence modeling.
- Relies heavily on quality and diversity of extracted features.

4.2.3 Summary: Classical Model Contribution

Classical ML models served as:

- Baseline performance benchmarks
- A way to test whether engineered features alone capture stress
  An early validation step before moving to costly deep learning training

They produced reasonable performance but struggled with:

- Sequential dependencies
- Long-term physiological drift
- Subtle stress transitions

Hence, deep learning architectures were explored next.

4.3 Deep Learning Architectures for Sequence Modeling

The second experimental objective was to capture temporal evolution of stress, something classical ML cannot do.
Five advanced 1D signal-sequence architectures were implemented and trained on an RTX 4060 GPU with the following optimizations:

- Mixed-precision training (FP16)
- Memory growth enabled
- Loss scaling for numerical stability

Deep models processed 10-window sequences, each representing 5 minutes × 10 continuous physiological segments.

4.3.1 CNN–BiLSTM–Attention Model

Working Principle

- 1D Convolutional Layers extract local temporal patterns (e.g., QRS shapes, EMG bursts).
- BiLSTM Layers learn long-term dependencies by processing the sequence in forward and backward directions.
- Attention Module assigns higher weights to physiologically important time segments.

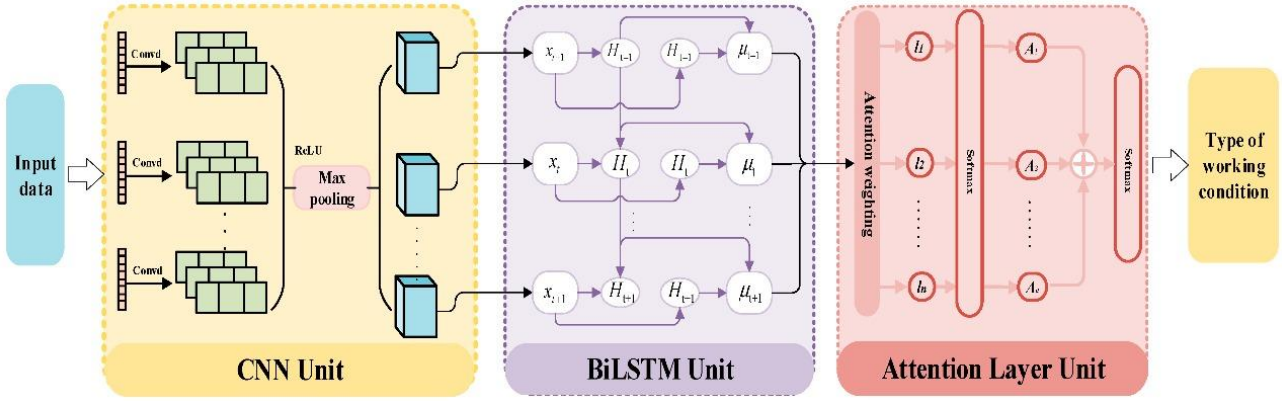


Fig 4.3 : CNN–BiLSTM–Attention Architecture Diagram

Mathematical Formulation

BiLSTM output:

$$h_t = [\overrightarrow{h_t}; \overleftarrow{h_t}] \tag{4.7}$$

Attention weight:

$$\alpha_t = \frac{\exp(W h_t)}{\sum_k \exp(W h_k)} \tag{4.8}$$

Final representation:

$$H = \sum_t \alpha_t h_t \tag{4.9}$$

Advantages

- Most accurate offline performance due to bidirectional processing.
- Strong ability to learn long-range dependencies.

Limitations

- Bidirectional processing is not ideal for real-time deployment, since future context is unavailable.
- Computationally heavier

4.3.2 Parallel CNN–LSTM

The Parallel CNN–LSTM model processes the input sequence through two convolutional streams, each using a different kernel size to capture complementary temporal patterns.

Architecture Summary

Dual CNN Streams:

- Kernel = 3: Learns short-term variations (fast ECG changes, EMG bursts, GSR spikes).
- Kernel = 5: Learns broader trends (slow HRV drift, respiration patterns, gradual stress buildup).

Each stream extracts features at a different temporal scale.

LSTM Layers:

The outputs of the two CNN streams are passed into separate LSTM layers, allowing the model to learn how each scale of features evolves over time.

Feature Fusion:

The two LSTM outputs are concatenated:

$$h_{\text{fusion}} = [h_1 \; ; \; h_2] \tag{4.10}$$

and fed into dense layers for final classification.

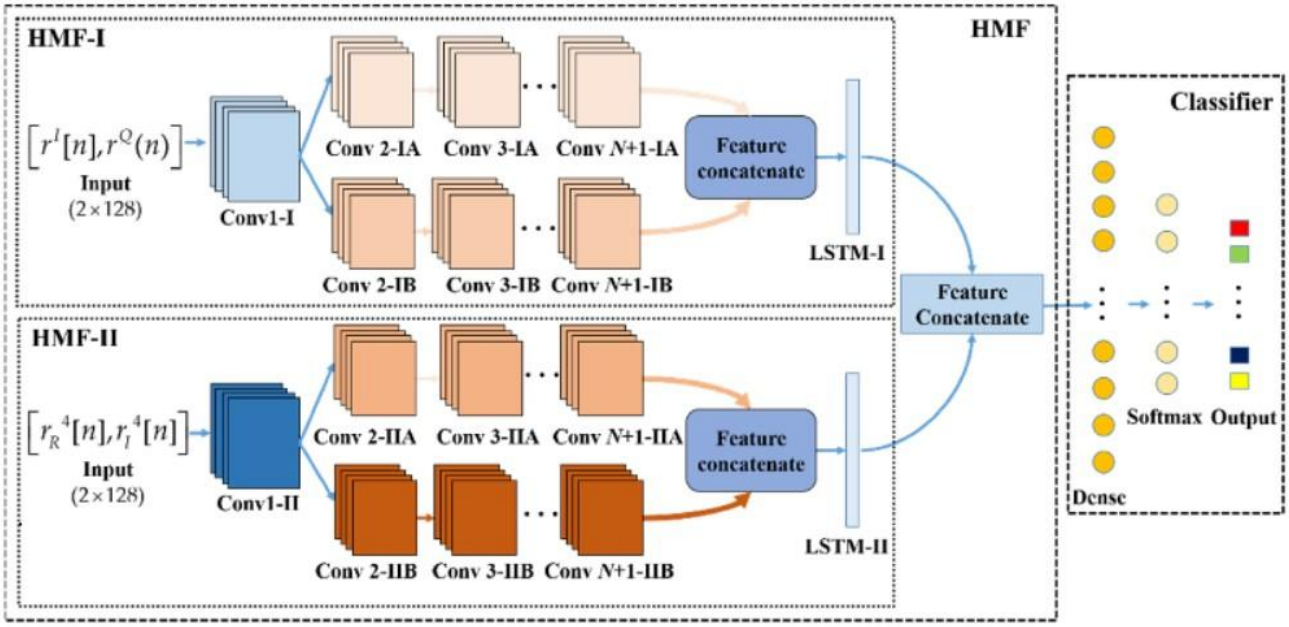


Fig 4.4 : Parallel CNN-LSTM Multi-Stream Architecture Diagram

Advantages

- Multi-scale feature learning (fine + coarse temporal patterns).
- Higher accuracy than single-branch CNN–LSTM.
- Moderate computational cost.

Limitations

- Heavier than GRU
- Lacks explicit attention mechanism

4.3.3 ResNet1D–LSTM

The ResNet1D–LSTM model integrates residual convolutional blocks with temporal sequence modeling. It is designed to extract deep hierarchical patterns from physiological signals while maintaining stable gradient flow.

Architecture Overview

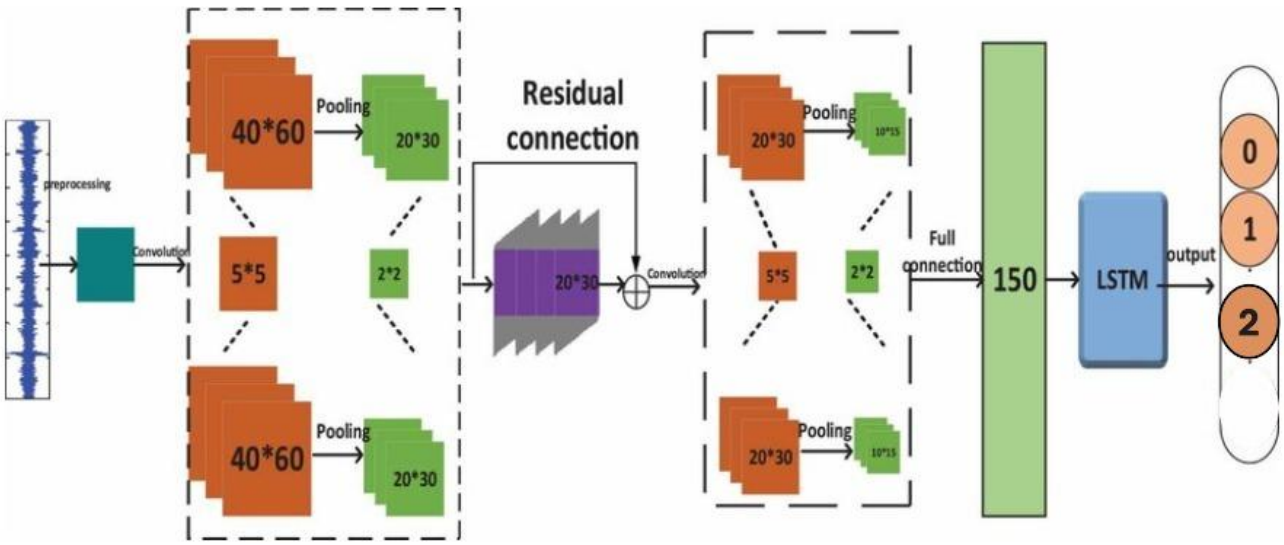


Fig 4.5 : ResNet1D-LSTM Architecture Diagram

Residual Convolution Blocks

Each block applies two 1D convolutions and adds the input back to the output:

$$y = F(x) + x \quad (4.11)$$

This skip connection prevents degradation in deeper networks and improves robustness to noisy multimodal signals.

LSTM Layers

After residual feature extraction, an LSTM layer models temporal dependencies across the sequence, enabling the detection of gradual stress accumulation and long-term variations in HRV, EMG, GSR, and respiration.

Advantages

- Extracts deep multi-scale temporal features effectively.
- Skip connections reduce vanishing gradient issues.

- Performs well on complex and noisy physiological data.

Limitations

- Higher parameter count than GRU models.
- Longer training time compared to lightweight architectures.

4.3.4 Inception1D–LSTM

The Inception1D–LSTM architecture extracts multi-scale temporal features by applying parallel convolutional operations before feeding the aggregated representation into an LSTM layer.

Architecture Overview

The Inception block applies four parallel transformations:

- **1×1 convolution** – dimensionality reduction and feature compression
- **3×3 convolution** – mid-scale temporal pattern extraction
- **5×5 convolution** – long-range temporal structure capture
- **Max-pooling pathway** – robust feature aggregation against noise

The outputs are concatenated and passed to an LSTM layer for temporal sequence modeling, enabling the network to learn both short- and long-duration stress signatures.

Advantages

- Captures multi-scale physiological stress patterns in a single block
- Strong performance under noisy or irregular sequences
- Effective for multimodal HRV–GSR–EMG signal fusion

Limitations

- Computationally heavier than GRU-based models

- Requires careful hyperparameter tuning for optimal performance

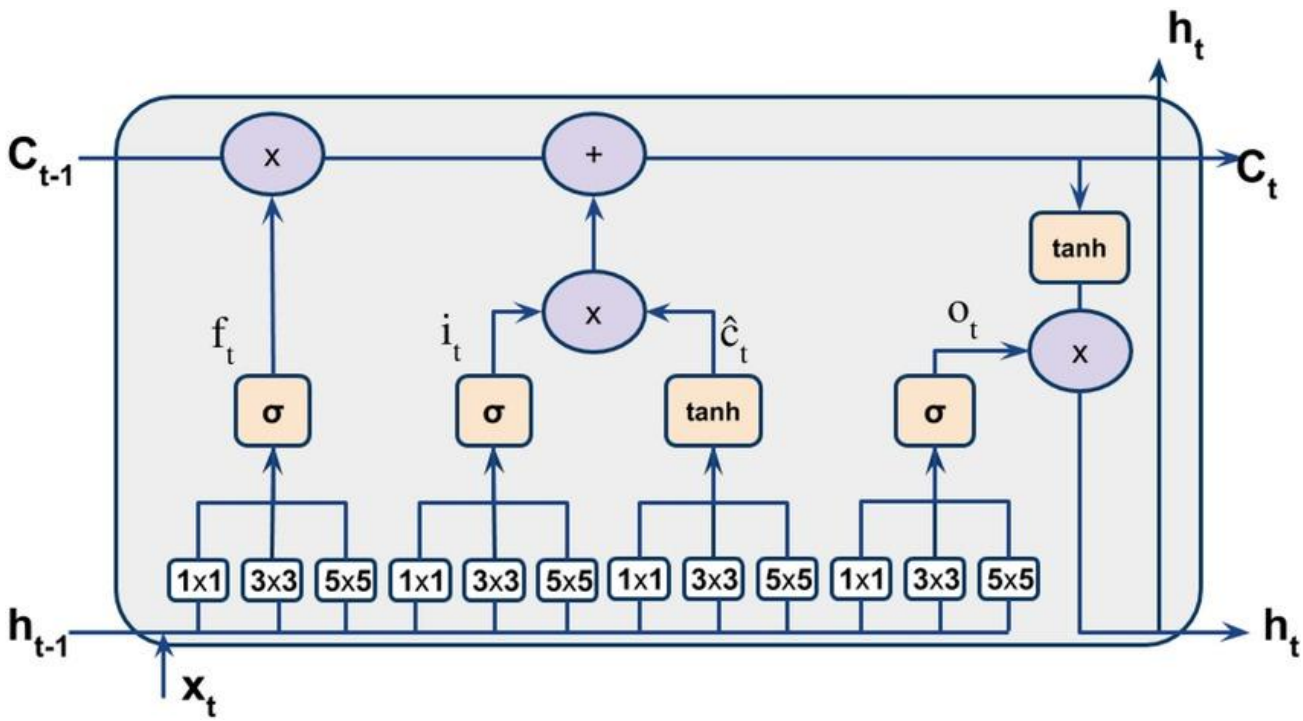


Fig 4.6: Inception1D-LSTM Architecture Diagram

4.3.5 CNN–GRU–Attention Model

The CNN–GRU–Attention architecture is a lightweight alternative to the BiLSTM model, designed specifically for efficient real-time stress detection.

Why GRU?

GRUs simplify the LSTM structure by merging gates, resulting in fewer parameters and faster computation.

A simplified GRU update equation is:

$$h_t = (1 - z_t)h_{t-1} + z_t\tilde{h}_t \quad (4.12)$$

where the update gate z_t controls how much past vs. new information is retained.

Architecture Overview

- **CNN layers** extract short-term temporal features (ECG morphology, micro-EMG bursts, GSR peaks).
- **GRU layers** model sequential dependencies using a compact recurrent unit.
- **Attention module** highlights important temporal regions relevant to stress changes.
- **Fully connected layers** map the attended sequence representation to low/medium/high stress classes.

The architecture is structurally identical to CNN–BiLSTM–Attention, except that LSTM layers are replaced with GRUs.

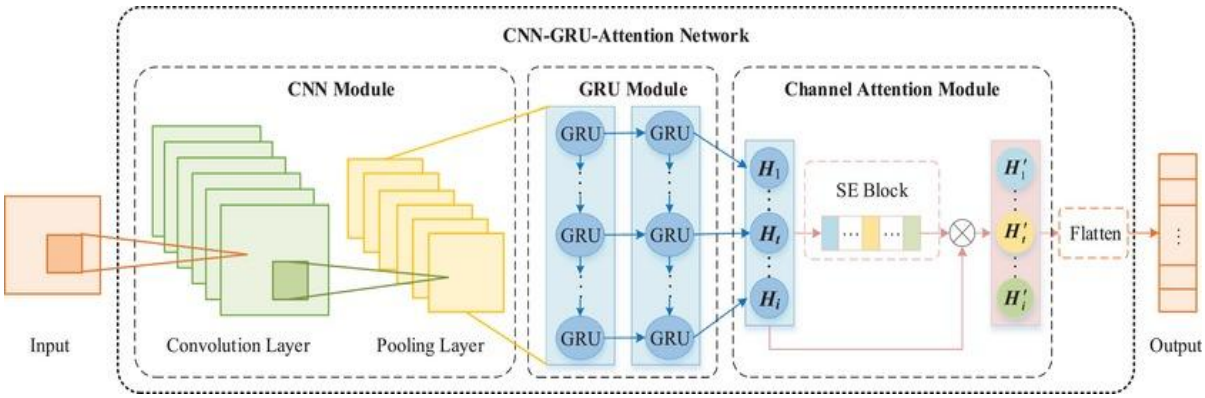


Fig 4.7: CNN-GRU-Attention Architecture Diagram

Advantages

- ~60% fewer parameters than BiLSTM
- Much faster inference → ideal for real-time, on-device deployment
- Requires less data to train effectively
- Lower memory footprint

Limitations

- Slightly weaker long-term temporal modeling
- Typically 2–3% lower accuracy compared to BiLSTM in offline evaluation

4.4 Rationale Behind Algorithm Selection

The system was developed following a structured, multi-stage approach, beginning with classical machine learning methods and evolving into advanced deep learning architectures. Each stage was chosen based on the nature of physiological signals, the need for temporal modeling, and real-time feasibility.

Phase 1: Classical Machine Learning Baseline

Classical ML models such as Random Forest, Gradient Boosting, and SVM were first explored using handcrafted features (HRV metrics, EMG statistical features, GSR tonic/phasic characteristics, and respiration variability).

Why Classical ML?

- These features are clinically interpretable and widely validated in stress research.
- Classical models provide strong baselines to benchmark improvement with deep learning.
- Training is computationally inexpensive, allowing rapid experimentation.

Process:

- Features were extracted from 5-minute windows across ECG, EMG, GSR, and respiration signals.
- Soft-voting and stacking ensembles were evaluated to improve generalization.

- Standardization, normalization, and feature scaling were performed consistently across all modalities.

Limitations:

- Classical ML cannot directly capture temporal evolution of stress.
- Feature quality heavily depends on preprocessing accuracy.
- Performance saturates when physiological patterns are subtle or overlapping.

Phase 2: Deep Learning for Sequential Modeling

After establishing baselines, the study progressed to sequence-based deep learning models that could analyze raw temporal patterns without handcrafted features.

Why Deep Learning?

Driving stress evolves gradually across time, often spanning multiple physiological cycles. Deep models can learn:

- long-range dependencies,
- multimodal correlations across signals,
- subtle temporal variations missed by static ML models.

Architectures explored included:

- CNN–BiLSTM–Attention (high accuracy, strong temporal modeling)
- Parallel CNN–LSTM (multi-scale feature extraction)
- ResNet1D–LSTM (deep residual temporal modeling)
- Inception1D–LSTM (multi-branch temporal filters)
- CNN–GRU–Attention (lightweight real-time model)

Phase 3: BiLSTM vs. GRU Selection

A deeper comparison was made between BiLSTM and GRU architectures based on accuracy, temporal modeling capability, computational cost, and real-time practicality.

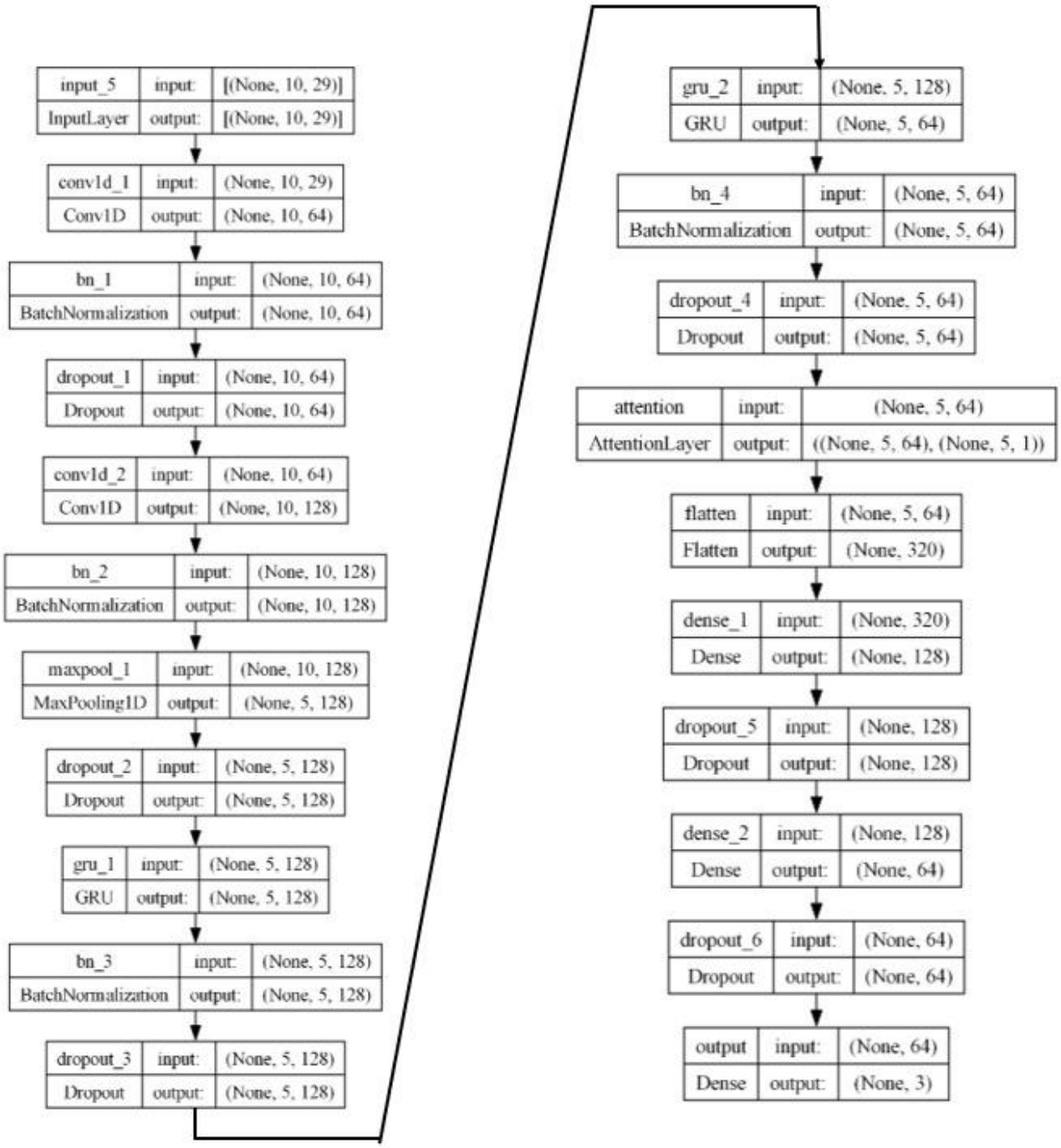
BiLSTM – Best for Offline Evaluation

- Captures forward and backward dependencies.
- Achieved the highest accuracy and F1 scores in controlled offline experiments.
- Strong for analyzing long-duration physiological sequences.

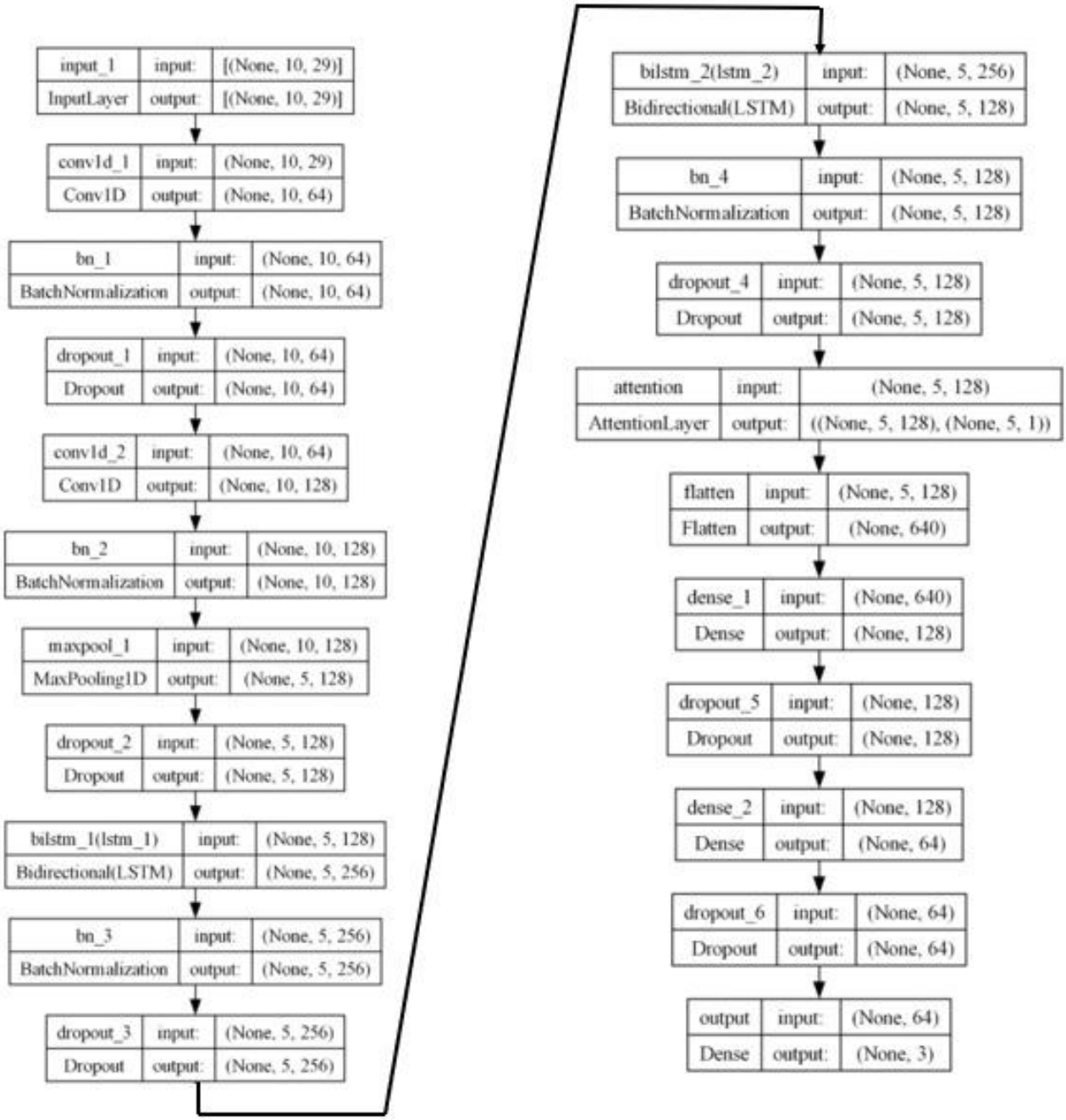
GRU – Best for Real-Time Deployment

- Forward-only processing avoids reliance on future context, which is unavailable in live settings.
- 60% fewer parameters → lower latency, faster inference.
- Suitable for on-board computation in vehicles.

This dual strategy ensures maximum performance during offline evaluation and efficient behavior during real-world deployment.



4.8 CNN-GRU-Attention Layer Architecture Diagram



4.9 CNN-BiLSTM-Attention Layer Architecture Diagram

4.7 Summary

The chapter presented a structured progression from handcrafted-feature classical ML models to advanced deep neural architectures optimized for temporal stress modeling. Key takeaways include:

- Classical ML models provided interpretability and served as strong baselines but were limited by their inability to model temporal patterns.
- Deep learning models, particularly CNN–BiLSTM–Attention and CNN–GRU–Attention, successfully leveraged multimodal temporal dependencies inherent in ECG, EMG, GSR, and respiratory signals.
- BiLSTM demonstrated superior offline accuracy, making it ideal for batch analysis.
- GRU emerged as the optimal candidate for real-time stress detection due to its efficiency and reduced parameter footprint.
- The algorithm selection was guided by application needs: accuracy for offline evaluation and speed for real-time driving scenarios.

The next chapter presents detailed experimental evaluations, confusion matrices, model performance comparisons, and insights derived from multimodal stress classification results.

CHAPTER 5

RESULTS AND DISCUSSIONS

This section presents the qualitative and quantitative results obtained from the multimodal driver stress detection system. Each objective is evaluated through multiple phases, covering classical machine learning baselines, deep learning architectures, and real-time–oriented models. Comparative metrics and model performance insights are provided to illustrate the effectiveness of each approach.

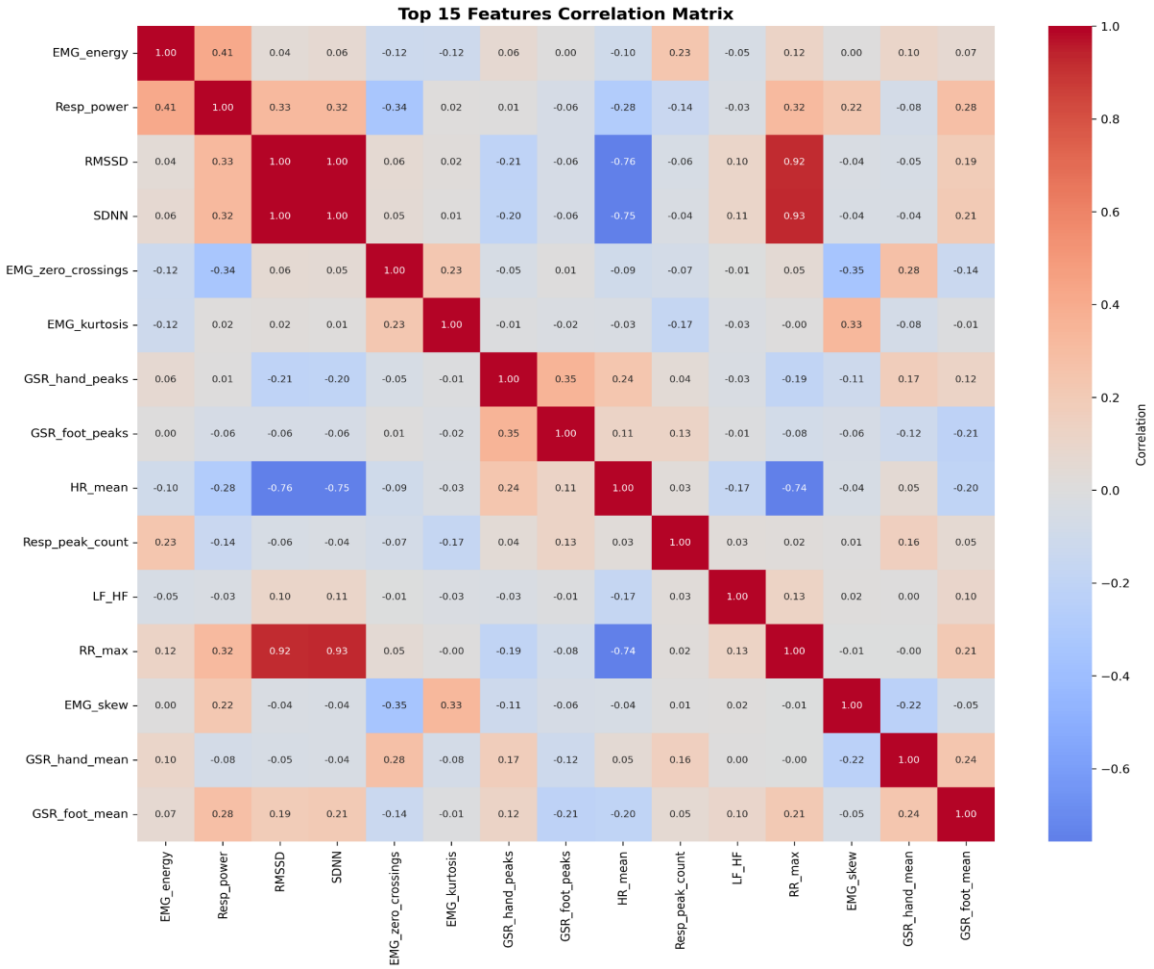


Fig 5.1: Top 15 Feature Correlation Matrix Heatmap

Objective 1: Classical Machine Learning for Stress Classification

Phase 1: Soft-Voting Ensemble (RF + GB + SVM)

The first phase evaluated a soft-voting ensemble combining Random Forest, Gradient Boosting, and RBF-SVM classifiers. The model used handcrafted physiological features such as HRV statistics, EMG energy descriptors, GSR tonic/phasic markers, and respiration variability.

The ensemble approach improved stability over single models and achieved balanced classification across low, medium, and high stress levels. However, it remained sensitive to overlapping physiological patterns and could not capture long-term temporal dynamics.

Phase 2: Stacking Ensemble with Gradient-Boosting Learners

To further enhance feature-level classification, a stacking ensemble was implemented using XGBoost and LightGBM as base learners with Logistic Regression as the meta-classifier. This method improved decision boundaries and reduced bias in medium-stress recognition.

While the performance improved, handcrafted features reached their representational limit, and the model could not distinguish subtle stress transitions.

Objective 2: Deep Learning for Sequential Stress Modeling

Phase 1: CNN–BiLSTM–Attention

This architecture achieved the best offline performance by capturing both short-term variations (via CNN) and long-range temporal trends (via BiLSTM).
The attention mechanism emphasized physiologically significant periods such as HRV drops, EMG bursts, and sudden GSR changes.

It demonstrated high generalization across subjects but required significant computation.

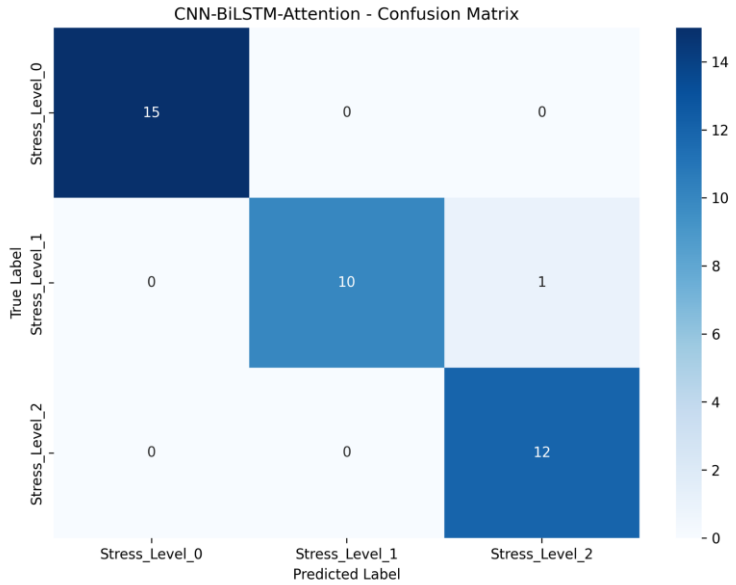


Fig 5.2: Confusion Matrix for CNN-BiLSTM-Attention Model

Phase 2: Parallel CNN–LSTM

Two parallel convolution branches with different kernel sizes learned multi-scale temporal features.
The combined representation improved medium-stress classification and reduced

noise sensitivity, but the model remained heavier than GRU-based variants.

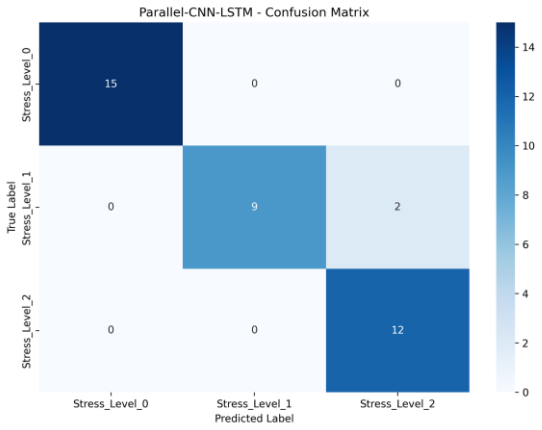


Fig 5.3: Confusion Matrix for Parallel CNN-LSTM Model

Phase 3: ResNet1D–LSTM

Residual connections stabilized deeper temporal processing, helping the model handle noisy multimodal signals.

Despite strong feature extraction, training time and parameter count were significantly higher.

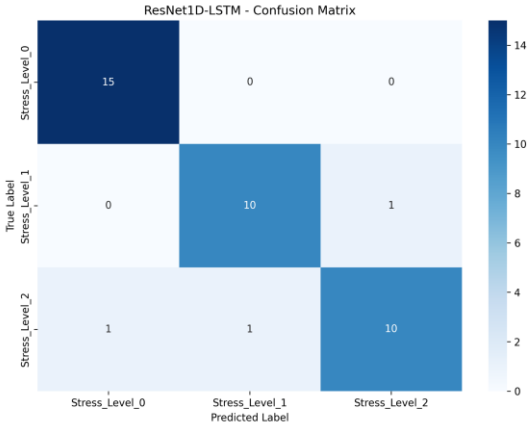


Fig 5.4: Confusion Matrix for ResNet1D-LSTM Model

Phase 4: Inception1D–LSTM

Parallel kernels (1×1, 3×3, 5×5) and max-pooling enabled multi-scale pattern recognition.

The model performed well but required extensive tuning to avoid overfitting.

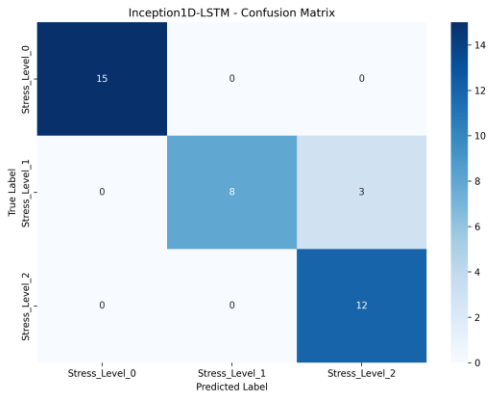


Fig 5.5: Confusion Matrix for Inception1D-LSTM Model

Phase 5: CNN–GRU–Attention

This model delivered the best balance between accuracy and efficiency. GRU's merged update/reset gates reduced computational load, making it ideal for real-time deployment while maintaining strong accuracy.

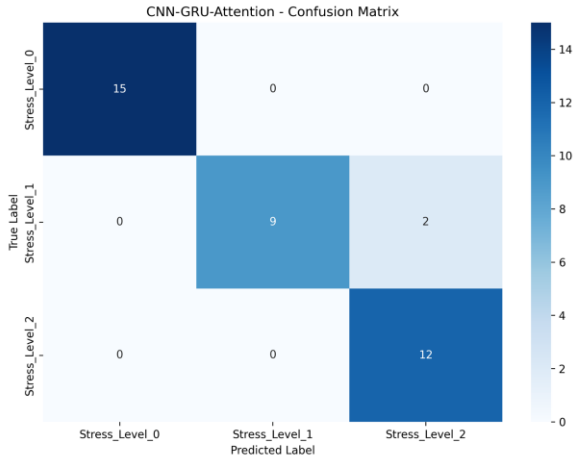


Fig 5.6: Confusion Matrix for CNN-GRU-Attention Model

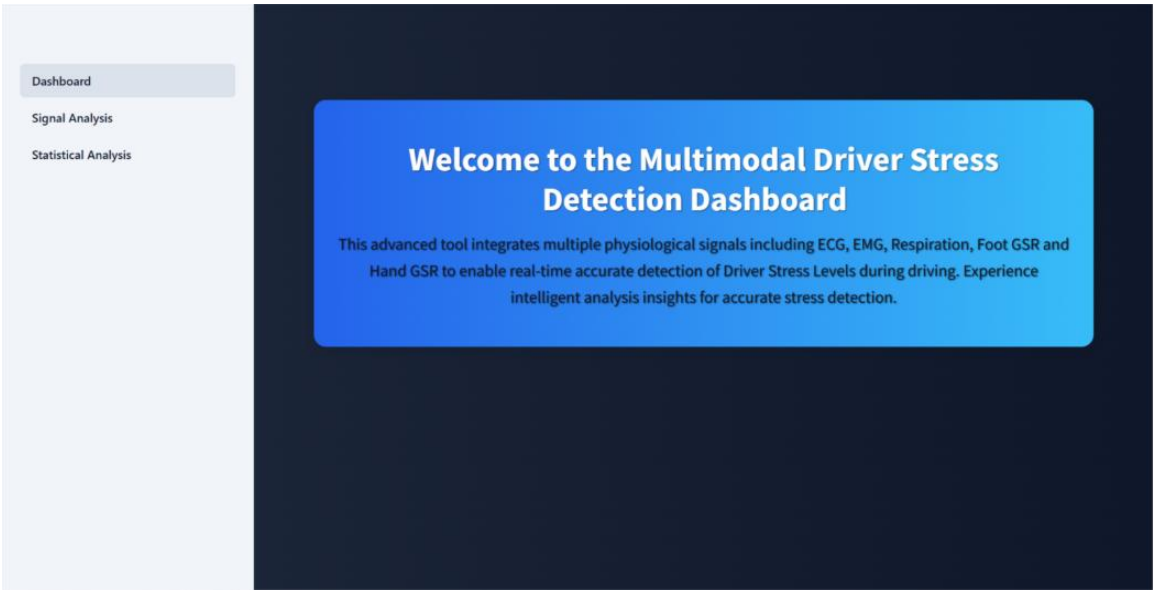
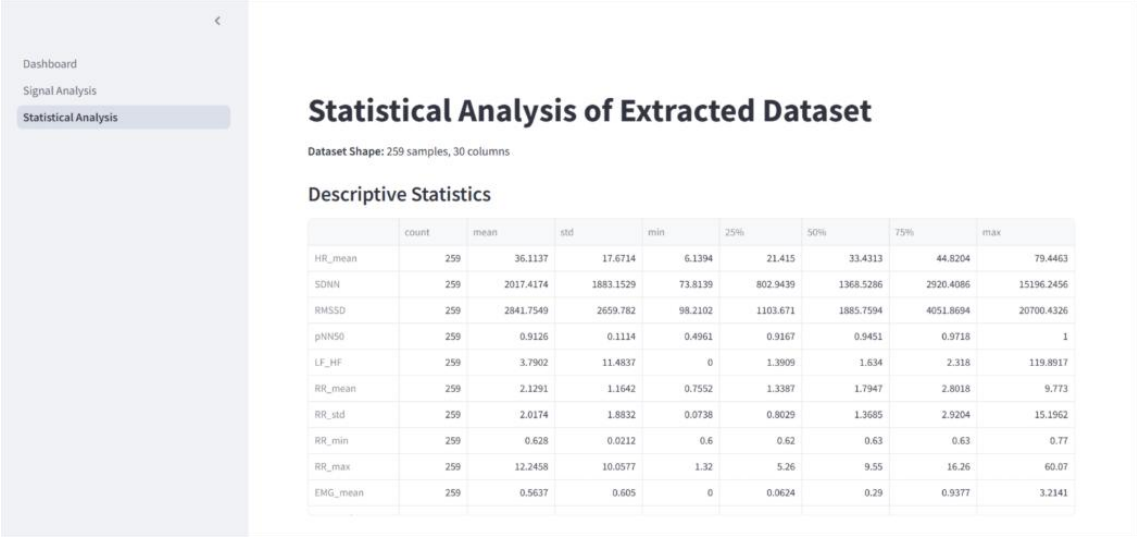


Fig 5.7 Website Dashboard



	count	mean	std	min	25%	50%	75%	max
HR_mean	259	36.1137	17.6714	6.1394	21.415	33.4313	44.8204	79.4463
SDNN	259	2017.4174	1883.1529	73.8139	802.9439	1368.5286	2920.4086	15196.2456
RMSSD	259	2841.7549	2659.782	98.2102	1103.671	1885.7594	4051.8694	20700.4326
pNN50	259	0.9126	0.1114	0.4961	0.9167	0.9451	0.9718	1
LF_HF	259	3.7902	11.4837	0	1.3909	1.634	2.318	119.8917
RR_mean	259	2.1291	1.1642	0.7552	1.3387	1.7947	2.8018	9.773
RR_std	259	2.0174	1.8832	0.0738	0.8029	1.3685	2.9204	15.1962
RR_min	259	0.628	0.0212	0.6	0.62	0.63	0.63	0.77
RR_max	259	12.2458	10.0577	1.32	5.26	9.55	16.26	60.07
EMG_mean	259	0.5637	0.605	0	0.0624	0.29	0.9377	3.2141

Fig 5.8 Statistical Analysis on the website

Objective 3: Deep Learning Model Comparison and Deployment Feasibility

This objective evaluates the performance of the two strongest deep learning architectures developed in this study which are CNN–BiLSTM–Attention and CNN–GRU–Attention. Both models were trained using multimodal sequences and assessed on accuracy, F1-score, parameter complexity, and real-time suitability.

The goal was to identify the model that offers the best balance between accuracy and computational efficiency for driver stress detection.

Phase 1: Performance Evaluation (Accuracy & F1-Score)

Both models were trained on the prepared multimodal dataset and evaluated using the same test split.

- **CNN–BiLSTM–Attention** achieved the highest overall accuracy and F1-score.
- **CNN–GRU–Attention**, while slightly behind in accuracy, provided strong and stable results.

Table 5.1 : Accuracy and F1 Scores comparison

Model	Accuracy	F1 Score
CNN-BiLSTM-Attention	0.9737	0.9736
CNN-GRU-Attention	0.9474	0.9468
ResNet1D-LSTM	0.9211	0.9198
Inception1D-LSTM	0.9211	0.9192
Parallel-CNN-LSTM	0.9362	0.9322

Phase 2: Computational Efficiency & Real-Time Suitability

This phase compared the models on:

- Parameter size
- Inference speed
- Suitability for in-vehicle, low-latency applications

The GRU-based model required 60% fewer parameters, making it more suitable for edge-device deployment and real-time inference.

Table 5.2 : Number of parameters comparison

Model	**Parameters**
CNN-BiLSTM-Attention	550,788
CNN-GRU-Attention	217,796
ResNet1D-LSTM	433,731
Inception1D-LSTM	349,891
Parallel-CNN-LSTM	336,899

CHAPTER 6
CONCLUSION

This study presents a multimodal framework for real-time driver stress detection using ECG, EMG, GSR, and respiration signals. It overcomes single-signal limitations by integrating handcrafted feature-based machine learning with deep learning models that capture temporal dynamics. The workflow evaluated classical models, hybrid convolution–recurrent networks, and attention-enhanced architectures.

Classical ML methods using HRV features, EMG statistics, respiration variability, and GSR tonic–phasic descriptors were tested. Soft-voting and stacking ensembles provided stable baselines and highlighted the interpretability of handcrafted indicators but struggled to track gradual physiological changes during real driving. The deep learning stage focused on models that learn temporal dependencies directly from multimodal sequences. Parallel CNN–LSTM, ResNet1D–LSTM, Inception1D–LSTM, CNN–GRU–Attention, and CNN–BiLSTM–Attention were implemented. The CNN–BiLSTM–Attention model performed best, benefiting from convolution-based pattern extraction, bidirectional temporal modeling, and attention to stress-relevant time steps.

The CNN–GRU–Attention model offered a lightweight, faster alternative with fewer parameters, making it suitable for real-time vehicular deployment despite slightly lower accuracy. This highlights a trade-off: BiLSTM models suit offline high-precision analysis, while GRU-based models are preferable for embedded, latency-sensitive environments. Overall, the framework shows that multimodal, sequence-aware physiological modeling enables reliable, high-accuracy driver stress detection with real-time feasibility, supporting next-generation intelligent monitoring systems.

CHAPTER 7

FUTURE WORKS

Several enhancements can further strengthen the multimodal driver stress detection system and broaden its real-world applicability. A major direction is developing an adaptive alerting mechanism that responds automatically to rising stress patterns, enabling timely notifications and reducing dependence on manual monitoring.

1. Adaptive Decision-Making Using Fuzzy Logic

Future versions can incorporate fuzzy inference rules to better interpret borderline or ambiguous stress levels. Since physiological responses rarely fit strict categories, fuzzy logic can smooth transitions, reduce false alarms, and support more human-like real-time decisions.

2. Multimodal Transformer-Based Fusion

Transformer architectures can be explored to model long-range dependencies across ECG, EMG, GSR, and respiration simultaneously. Cross-modal attention would capture richer physiological interactions, improving robustness and classification accuracy in noisy driving environments.

3. Real-Time Conversational AI Assistance

A major future direction is integrating a conversational AI feedback module that interacts with the driver during stressful conditions. This voice-based assistant could

- provide calming prompts,
- suggest breathing exercises,
- warn the driver when stress spikes sharply, and
- offer supportive dialogue to maintain alertness.

Such a system shifts the framework from passive monitoring to active driver support, enabling a more personalized and safety-focused interaction.

4. Expanded Dataset and Context-Aware Modeling

Future work may include collecting diverse driving data covering night-time conditions, heavy traffic, fatigue, and emotional disturbances. Integrating contextual cues such as speed, steering behavior, and lane deviation could create a more holistic driver-state monitoring system.

5. Real-Time Deployment Optimizations

Techniques such as model pruning, quantization, and edge-device optimization can enhance performance for in-vehicle deployment. These improvements enable low-latency inference suitable for onboard automotive hardware.

Overall, these advancements have the potential to transform the proposed framework into a fully adaptive, intelligent, and interactive driver assistance system, significantly improving road safety and user experience.

CHAPTER 8

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